CS4200 Semester Project

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ISA Extension of RV32I

- Added M-type instructions
- Mul
- Mulh
- Div
- Mod
- Manual calculations using the algorithms from lecture slides, instead of using built in Python functions

What Was Modified

- alu_exec added implementation of each M-type instruction calculation
- try_mnemonic for logging/testing
- write_lines, trace_instruction, and main now create an additional output file to show instruction types explicitly

```
# ------------------------------------
# Additional Instructions for Project
# ------------------------------------
if f7 == 0b0000001:
    if f3 == 0b000:
        return "mul"
    if f3 == 0b001:
        return "mulh"
    if f3 == 0b100:
        return "div"
    if f3 == 0b110:
        return "mod"
```

```
def trace_instruction(id_out):
    d = id_out["d"]
    mnem = try_mnemonic(d)
    string = "0x%08X: %s" % (d["instr"], mnem)

    if mnem in ["mul", "mulh", "div", "mod"]:
        string += " <-- New instruction"

    return string
```

Additional Files

- Used AI to create a comprehensive input file for extensive testing and demonstration
- Used dmem_final to check all calculations of M-type instructions

```
# pc 0x0000008C: mul x20,x3,x3 ; expected mem[264] = 25, -5*-5 low
02318A33
# pc 0x00000090: sw x20,8(x10) ; expected mem[264] = 25, -5*-5 low
01452423
# pc 0x00000094: mul x20,x6,x4 ; expected mem[268] = 0, low half overflow
02430A33
# pc 0x00000098: sw x20,12(x10) ; expected mem[268] = 0, low half overflow
01452623
# pc 0x0000009C: mulh x20,x6,x4 ; expected mem[272] = 1, high half positive overflow
02431A33
# pc 0x000000A0: sw x20,16(x10) ; expected mem[272] = 1, high half positive overflow
01452823
# pc 0x000000A4: mulh x20,x3,x1 ; expected mem[276] = 0xFFFFFFFF, -5*7 high
02119A33
```

```
0x00000100 : 0x00000015 (21)          # mul 7 * 3
0x00000104 : 0xFFFFFFDD (-35)         # mul 7 * -5 low 32 bits
0x00000108 : 0x00000019 (25)          # mul -5 * -5
0x0000010C : 0x00000000 (0)           # mul 0x40000000 * 4 low 32 bits
0x00000110 : 0x00000001 (1)           # mulh 0x40000000 * 4
```

Algorithms + Demonstration